

Could the Ban on Improper Movement be about binding after all?*

Peter Grishin

Brown University

1. Introduction

Chomsky, in “Conditions on Transformations”, proposed that an XP cannot move from Spec,CP to non-Spec,CP positions (a “modernized” restatement of Chomsky 1973:257, ex.123c). This was motivated, in large part, by the unavailability of hyperraising in English. May (1979) offered an influential binding-theoretic explanation for this that had the more general consequence of blocking movement from any $\bar{A}$ position to any A position. May’s account (and related ones) made crucial use of the different binding-theoretic status of A vs. $\bar{A}$ traces in the theory of the time (e.g. May 1979, Chomsky 1981, 1986, a.o.). Later, Chomsky (1981:195,199,203) called such movements “improper”, and since then “improper movement” has commonly been understood to refer to movement from an $\bar{A}$ position to an A position,¹ and the “Ban on Improper Movement” (BoIM) has become the term for the ban on such movement dependencies:

- (1) **Ban on Improper Movement (BoIM):** $\bar{A}$ -movement cannot feed A-movement.

With the advent of Minimalism, these kinds of approaches fell out of fashion. Most recent analysis of improper movement effects instead seek to explain (something like) one of the following broader proposed generalizations, usually without appealing to binding properties:

*For useful discussion of, questions about, and productive disagreement on the topics, data, and ideas in this talk, I’d like to thank Elena Benedicto, Tanya Bondarenko, Amy Rose Deal, János Egressy, Matthew Hewett, Polly Jacobson, Kyle Johnson, Filipe Hisao Kobayashi, Magdalena Lohninger, Tom Meadows, Line Mikkelsen, Norvin Richards, Will Oxford, David Pesetsky, the audience and reviewers of CLS 61 and NELS 56, participants in the Fall 2024 *Syntax of a Language (Family)* seminar at MIT, and members of the MIT Passamaquoddy group. I’d also like to thank the speakers of Passamaquoddy-Wolastoqey that I’ve been lucky enough to work with over the years, without whom none of this would be possible—*psi-te kiluwaw* ‘koti-wolasuweltomulpa, *pciliw* Margaret Apt, Edwina Mitchell, Gracie Paul, naka Roger Paul, *eli ahtolokehkimiyeq nuli-*’sotomon ktolatuwewakonuwa.

¹Though not by everyone, e.g. Keine (2018), among others.

- (2) **Generalized Ban on Improper Movement (GBoIM):**² A phrase cannot move from a position X to a position Y where Y is lower on the functional sequence than X.
(e.g. Williams 2003, 2011, 2013, Müller 2014, Keine 2019, 2020, Poole 2022, Bondarenko 2024, Meadows 2024, Messick and Raghotham 2024; with important precursors in Williams 1974, Van Riemsdijk and Williams 1981).
- (3) **Generalized A-after- $\bar{A}$ Constraint (GAa $\bar{A}$ C):** A DP that has $\bar{A}$ -moved cannot then participate in any kind of A-dependency: A-movement, ϕ -agreement, case-assignment (particularly natural if all A-relations are driven by Agree with an A feature).
(e.g. Rezac 2003, Obata and Epstein 2011, Safir 2019, Wurmbrand 2019, Lohninger et al. 2022, Hewett 2025a)

For whatever reason, it seems less popular nowadays to try to derive and explain the “classic” Ban on Improper Movement (1) (though see, e.g., Abels 2007, 2009, Neeleman and Van De Koot 2010), much less by appealing to binding.

In this short paper I’d like to vindicate both the classic Ban on Improper Movement as well as a binding-based account of it. I show that Sauerland (1998, 2004), Ruys (2000), and van Urk’s (2015) approach to the distinct binding behavior of A and $\bar{A}$ movement (e.g. with regards to weak crossover (WCO), depictives, and parasitic gaps), in which A movement involves binding individual (type *e*) variables and $\bar{A}$ movement involves binding choice function variables (type (*et*)*e*), automatically derives the BoIM: following van Urk’s (2015) rules for interpreting A vs. $\bar{A}$ movement at LF, $\bar{A}$ movement followed by A movement results in a semantically uninterpretable structure, with irreparable type clashes. I then argue that this approach is superior to alternative approaches that instead try to derive the GBoIM or GAa $\bar{A}$ C, as those generalizations are too strong and underpredict attested types of cross-clausal A-dependencies crosslinguistically (cf. Grishin 2025, Mikkelsen et al. 2026).

2. Interpreting A and $\bar{A}$ chains: WCO, depictives, and parasitic gaps

A well-known difference between A and $\bar{A}$ movement is that only $\bar{A}$ movement is subject to weak crossover (WCO) effects:

- (4) WCO effects with $\bar{A}$ movement *English (Germanic)*
 a. ***Which final paper_i** did **its_i** author give up on **t_i**?
 b. ***Which final paper_i** did it seem to **its_i** author [**t_i** to be unsalvageable]?
- (5) No WCO effects with A movement *English (Germanic)*
 a. **No final paper_i** was given up on **t_i** by **its_i** author.
 b. **No final paper_i** seemed to **its_i** author [**t_i** to be unsalvageable].

²Also often known as the “Williams Cycle”.

Could the BoIM be about binding after all?

Given the following standard assumptions (e.g. Heim and Kratzer 1998, a.m.o.), this is very puzzling:

1. All movement involves inserting at LF lambda binders over variables of type *e* below a landing site of movement.
2. Traces (of DP movement) and pronouns are both interpreted as variables of type *e*.

It seems, instead, that we need some kind of *semantic* difference between A and $\bar{A}$ movement, related to binding, in order to capture this contrast.

Evidence for this comes from the observation that WCO also arises with base-generated $\bar{A}$ dependencies, like island-insensitive resumption (Hewett 2023, 2025b):

(6) Secondary WCO with island-insensitive resumption (Hewett 2025b:199–200)

a. *Iraqi Arabic (Semitic)*

*titðakkiri:n [firqat minu;]_k waʕadna ahl-a; innu raḥ nismaʕ
remember.2FSG band who promised.1PL family-his that FUT listen.to.1PL
ajj uynijja [isld. hijja_k titʕalliʕ-ha]?
any song it put.out.3FSG-it
(lit.) ‘Do you remember [whose; band]_k we promised his; family that we
would listen to any song [isld. it_k puts out]?’

b. *Syrian Arabic (Semitic)*

*btitzakkiri [zo:ʒ ajja binit;]_k ma kənti ḥardʕiri l-yom
remember.2FSG husband which girl NEG were.2FSG present.FSG the-day
[isld. lli ʕarrafnā ixwāt-ha; ʕale:-Ø_k fi:-Ø]?
that introduced.1PL siblings-her; to-him in-it
(lit.) ‘Do you remember [which girl’s; husband]_k you were absent the day
[isld. that we introduced her; siblings to him_k]?’

Here, the *wh* phrases *firqat minu* ‘whose band’ and *zo:ʒ ajja binit* ‘which girl’s husband’ are base-generated in Spec,CP, binding resumptive pronouns inside relative clause islands. Despite this, the possessors *minu* ‘who’ and *ajja binit* ‘which girl’ can’t bind pronouns c-commanding the resumptive, whether that pronoun is outside the island (6a) or inside (6b)³. In other words, WCO is not (just) about $\bar{A}$ movement, but about $\bar{A}$ binding, which is involved in $\bar{A}$ movement, but also shows up in other situations as well.

Other differences between A and $\bar{A}$ movement include the fact that A (but not $\bar{A}$) movement licenses depictives (7–8), and $\bar{A}$ (but not A) movement licenses parasitic gaps (9–10):

(7) Depictive licensing with A movement *English (Germanic)*

- a. Noor_i was given *t_i* that information drunk_i.
- b. Ethan_i was bumped into *t_i* shirtless_i.

³Possessors can bind out of DPs as long as WCO isn’t violated; see Hewett (2023, 2025b).

- (8) No depictive licensing with $\bar{A}$ movement *English (Germanic)*
 a. ***Who_i** did you give **t_i** that information **drunk_i**?
 b. ***Who_i** did you bump into **t_i shirtless_i**?
- (9) Parasitic gap licensing with $\bar{A}$ movement *English (Germanic)*
 a. **Which manuscript_i** did you submit **t_i** [after carefully rewriting **pg_i**]?
 b. **These potatoes_i** I picked up **t_i** yesterday [only to throw away **pg_i** today].
- (10) No parasitic gap licensing with A movement *English (Germanic)*
 a. ***My manuscript_i** was submitted **t_i** [after carefully rewriting **pg_i**].
 b. ***These potatoes_i** were picked up **t_i** yesterday [only to throw away **pg_i** today].

Similarly to WCO, these contrasts are commonly explained as arising through the different binding properties of A and $\bar{A}$ movement: only the kind of binding that A movement induces can bind the individual argument variable of a depictive secondary predicate (cf. Pylkkänen 2008, a.m.o.), and only the kind of binding that $\bar{A}$ movement induces can bind parasitic gaps (cf. Nissenbaum 1998, a.m.o.).

Van Urk (2015), Sauerland (1998, 2004) and Ruys (2000) propose that A movement involves abstraction over individuals (type e) whereas $\bar{A}$ movement involves abstraction over choice functions (type $(et)e$):

- (11) $\begin{array}{c} \text{A} \\ \swarrow \quad \searrow \\ \text{XP} \boxed{\lambda x} \quad \dots t \dots \end{array}$ (12) $\begin{array}{c} \bar{\text{A}} \\ \swarrow \quad \searrow \\ \text{XP} \boxed{\lambda f} \quad \dots t \dots \end{array}$

An important assumption here is a choice function analysis of quantifiers (type $((et)e)tt$, quantifying over choice functions; e.g. Sauerland 1998, 2004, Abels and Martí 2010):

- (13) Schema for quantifier denotations: $\lambda R_{((et)e)t}. Qf[R(f) = 1]$, where $Q \in \{\exists, \forall, \dots\}$

Van Urk's (2015) implementation of this analysis involves the following rules that apply at LF, after spellout, to convert (narrow-)syntactic representations into LF-interpretable ones:

- (14) *Interpreting A chains*
- a. **Interpretation rule for A chains:** In a movement structure formed by Agree for ϕ -features, adjoin a node λx to the probing head. (van Urk 2015:41)
- b. **Trace Conversion:** (Fox 2002:67 in van Urk 2015:41)
Variable Insertion: (Det) Pred $\rightarrow$ (Det) [Pred $\lambda y(y = x)$]
Determiner Replacement: (Det) [Pred $\lambda y(y = x)$] $\rightarrow$ the [Pred $\lambda y(y = x)$]

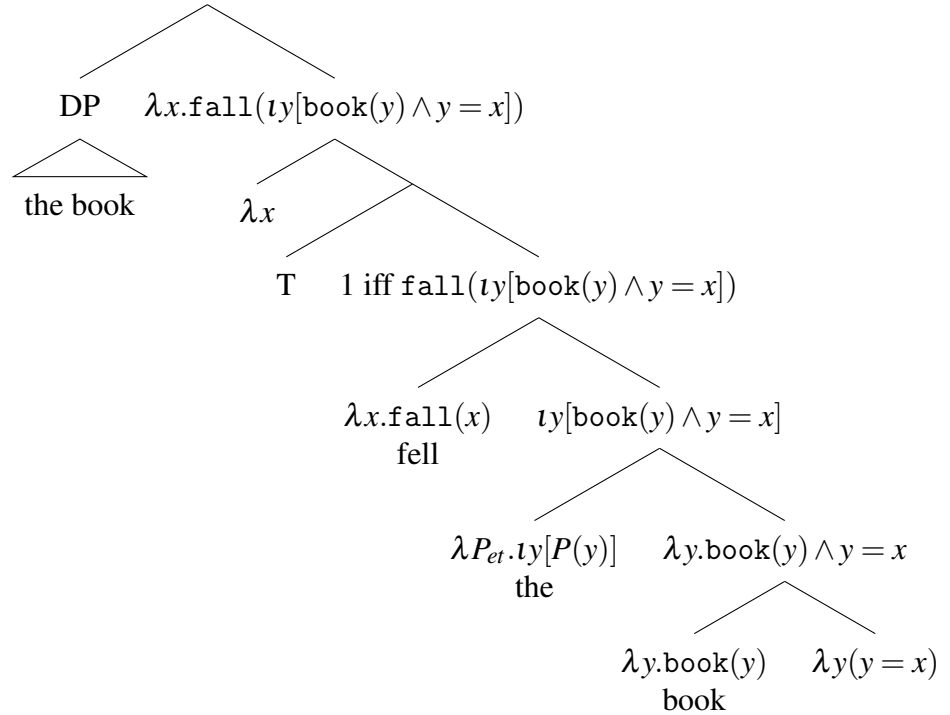
(15) *Interpreting $\bar{A}$ chains*

- a. **Interpretation rule for $\bar{A}$ chains:** In a movement structure formed by Agree for an $\bar{A}$ -feature, adjoin a node λf to the probing head. (van Urk 2015:40)
- b. **Quantificational Copy Conversion:** (van Urk 2015:39)
Quantifier Replacement: Quant Pred $\rightarrow f$ Pred
- c. **LF distributed deletion:** The NP restrictor of moved quantificational DPs is deleted at LF if necessary for composition. (based on van Urk 2015:39–41)

Note that Trace Conversion and Quantificational Copy Conversion act on *lower* copies.

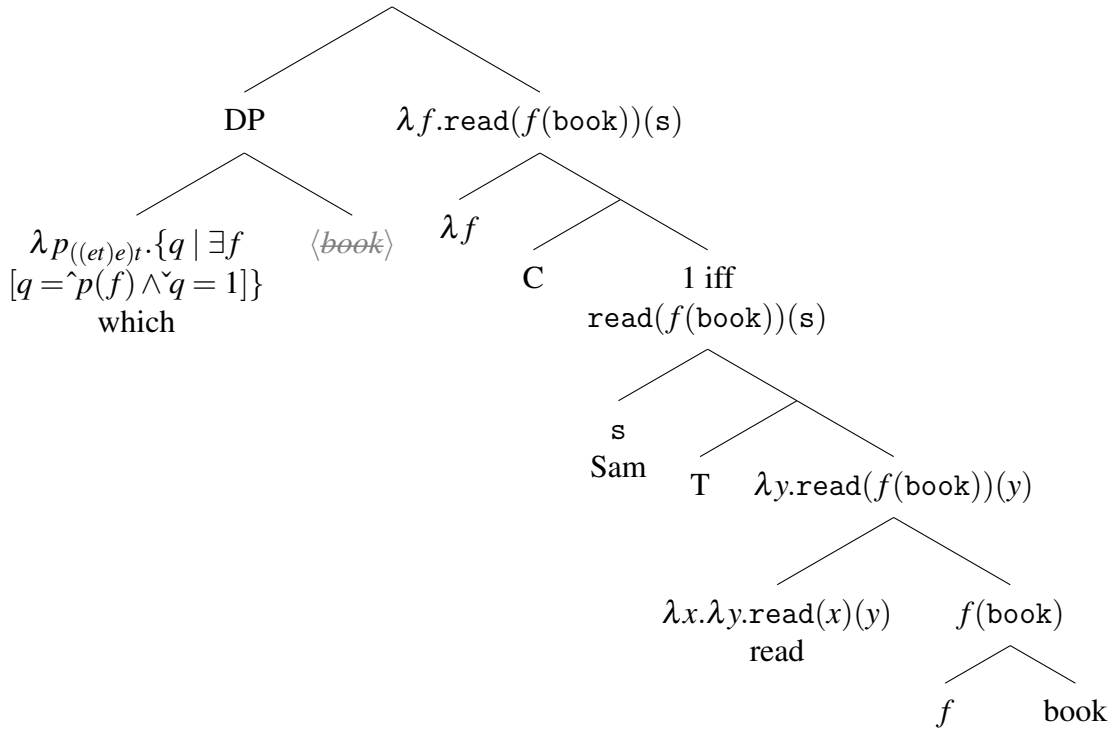
To illustrate the A chain interpretation rules, (16) provides a toy example of A movement, with all interpretation rules applied:

- (16) a. The book fell ~~the book~~.
- b. $1 \text{ iff fall}(\iota y[\text{book}(y) \wedge y = \iota x[\text{book}(x)])] \rightsquigarrow 1 \text{ iff fall}(\iota x[\text{book}(x)])$



To illustrate the $\bar{A}$ chain interpretation rules, (17) provides a toy example of $\bar{A}$ movement, with all interpretation rules applied (you can straightforwardly adapt this to your preferred semantics for questions; here I'm following Ruys 2000 for simplicity). For reasons of space, I do not illustrate how this approach derives differences between A and $\bar{A}$ movement with WCO, depictives, and parasitic gaps—see van Urk (2015) for details.

- (17) a. Which book did Sam read $\langle \text{which book} \rangle$?
 b. $\{q \mid \exists f[q = \hat{\text{read}}(f(\text{book}))(\text{s}) \wedge \check{q} = 1]\}$



3. Deriving the BoIM: $\bar{A}$ after A composes, A after $\bar{A}$ does not

Van Urk (2015) shows that this system allows for the structure generated by a DP A-moving and then $\bar{A}$ -moving to compose successfully. However, as far as I'm aware, it hasn't been noted that applying these rules to a reverse kind of derivation, in which a DP $\bar{A}$ -moves and then A-moves, results in a failure of composition due to type clashes, thus deriving the BoIM as a consequence of differences in the binding properties of A and $\bar{A}$ movement.

To illustrate the successful ordering, here's a schematic derivation of a DP containing a quantifier Q that first undergoes A movement and then $\bar{A}$ movement (based on van Urk 2015:41–42, ex.31):

- (18) $\bar{A}$ after A (“proper” movement): composes successfully

- a. $\overbrace{[\text{DP } Q \text{ NP}] \dots [\text{DP } Q \text{ NP}] \dots [\text{DP } Q \text{ NP}] \dots}^{\bar{A}}$ *Output of narrow syntax*
 b. $[\text{DP } Q \text{ NP}] \lambda f \dots [\text{DP } Q \text{ NP}] \lambda x \dots [\text{DP } Q \text{ NP}] \dots$ *Interpretation rules*
 c. $[\text{DP } Q \text{ NP}] \lambda f \dots [\text{DP } Q \text{ NP}] \lambda x \dots [\text{DP the } [\text{NP } \lambda y(y = x)]] \dots$ *Trace Conversion*
 d. $[\text{DP } Q \text{ NP}] \lambda f \dots [\text{DP } f \text{ NP}] \lambda x \dots [\text{DP the } [\text{NP } \lambda y(y = x)]] \dots$ *Quantificational Copy Conversion*
 e. $[\text{DP } Q \langle \text{NP} \rangle] \lambda f \dots [\text{DP } f \text{ NP}] \lambda x \dots [\text{DP the } [\text{NP } \lambda y(y = x)]] \dots$ *LF distributed deletion*

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In this derivation, the narrow syntax outputs a structure with (at least) three copies of DP (18a). We then insert the appropriate lambda binders (18b): λx for the lower step of A movement, and λf for the higher step of $\bar{A}$ -movement. Next, the corresponding lower-copy-conversion rules are applied: Trace Conversion for the lower copy of A-movement (18c), and Quantificational Copy Conversion for the lower copy of $\bar{A}$ movement (18d). Finally, we delete the NP restrictor of Q in the highest copy (18e). This results in an interpretable LF: $f(\llbracket \text{NP} \rrbracket^g)$ in the middle copy, of type e , can saturate its complement predicate of individuals (type et), and Q in the highest copy (type $((et)e)tt$) can compose with its complement predicate of choice functions (type $((et)e)t$).

In contrast, applying these movement types in reverse order results in type clashes:

(19) A after $\bar{A}$ (improper movement): **type clashes!**

- | | | |
|----|--|---|
| | $\begin{array}{c} \text{A} \qquad \qquad \bar{A} \\ \swarrow \quad \searrow \quad \swarrow \quad \searrow \\ \text{[DP Q NP]} \dots \text{[DP Q NP]} \dots \text{[DP Q NP]} \end{array}$ | |
| a. | $[\text{DP Q NP}] \dots [\text{DP Q NP}] \dots [\text{DP Q NP}]$ | <i>Output of narrow syntax</i> |
| b. | $[\text{DP Q NP}] \lambda x \dots [\text{DP Q NP}] \lambda f \dots [\text{DP Q NP}] \dots$ | <i>Interpretation rules</i> |
| c. | $[\text{DP Q NP}] \lambda x \dots [\text{DP Q NP}] \lambda f \dots [\text{DP } f \text{ NP}] \dots$ | <i>Quantificational Copy Conversion</i> |
| d. | $[\text{DP Q NP}] \lambda x \dots [\text{DP the } [\text{NP } \lambda y(y = x)]] \lambda f \dots [\text{DP } f \text{ NP}] \dots$ | <i>Trace Conversion</i> |
| e. | $[\text{DP Q } \langle \text{NP} \rangle] \lambda x \dots [\text{DP the } [\text{NP } \lambda y(y = x)]] \lambda f \dots [\text{DP } f \text{ NP}] \dots$ | <i>LF distributed deletion</i> ⁴ |

The narrow syntax outputs a structure with (at least) three copies of DP (19a). We then insert the appropriate lambda binders (19b): λf for the lower step of $\bar{A}$ movement, and λx for the higher step of A-movement. Next, the corresponding lower-copy-conversion rules are applied: Quantificational Copy Conversion for the lower copy of $\bar{A}$ movement (19c), and Trace Conversion for the lower copy of A-movement (19d). Finally, we delete the NP restrictor of Q in the highest copy (19e). The result of all this is an uninterpretable LF! The middle copy is of type e , which can't compose with its complement predicate of choice functions (type $((et)e)t$). The highest copy is a quantifier over choice functions (type $((et)e)tt$), which can't compose with its complement predicate of individuals (type et). We thus derive the BoIM as a consequence of the different ways that A and $\bar{A}$ chains are interpreted at LF.

⁴Applying LF distributed deletion to the intermediate copy would allow the intermediate step of movement to compose as long as *the* is of the right type (i.e. $((et)e)tt$ or $(et)e$) (indeed, the Trace Conversion rule assumes that the *the* that's inserted under Trace Conversion is of type $(et)e$). The problem with doing this is that then λx wouldn't be binding anything, because we've deleted the only expression containing the variable x , and so the derivation wouldn't be able to compose properly for other reasons. As for the higher copy, failing to apply LF distributed deletion to the highest copy wouldn't help things because Q (type $((et)e)tt$) is not the right type to compose with its NP restrictor (type et). Therefore, failures of composition are unavoidable.

4.1 Contra the Generalized Ban on Improper Movement

(20) a. *Passamaquoddy-Wolastoqey (Eastern Algonquian)*
 N-pawatom-**uw-a**-n-Ø **Piyel** [*⟨Piyel⟩* 'sukolopan-ka-n].
 1-want.TI-**APPL-3OBJ**-SG-IN.SG **Peter** 3-cake-make.AI-SG
 'I want **Peter** to make a cake.' (GP, MA 2023.05.02)

b. *Kalaallisut (Inuit)* (Mikkelsen and Thrane 2024:299)
Illit eqqama-**vakkit** [*⟨illit⟩* angerla-jaar-tutit].
2SG.ABS remember-**1SG:2SG.IND** leave-early-2SG.PTCP
 'I remember that **you** left early.'

Proponents of the GBoIM try to argue their way out of these kinds of examples. For instance, Poole (2022) proposes to explain away certain instances of RtO as actually involving ECM, without cross-clausal A movement. However, this approach fails to generalize, since at least some instances of RtO really cannot receive an ECM analysis, like in Passamaquoddy-Wolastoqey and Kalaallisut (see citations above). Taking another tack, Meadows (2024) proposes that movement driven purely by considerations of PF or LF interpretability (and not by (narrow-)syntactic features) is exempt from the GBoIM, arguing that genuine instances of RtO are not feature-driven. The intuition is that the GBoIM is a constraint on syntactic structure building, and thus doesn't apply to PF or LF operations.

⁵Some others have gone the other way and argued that the BoIM is actually too *strong* and undergenerates, e.g. Kobayashi (2020).

Could the BoIM be about binding after all?

The problem with this approach is that a PF/LF-interpretability account of some instances of RtO is highly implausible. For Passamaquoddy-Wolastoqey, it's unclear what the PF requirement would be, because RtO can be both overt and covert (likewise in Kalaallisut; Mikkelsen and Thrane 2024):

- (21) Overt and covert RtO available *Passamaquoddy-Wolastoqey (E. Algonquian)*
- a. Piyel **w-ikuwoss-ol** Ø-nokatom-**uw-a-n** [Ø-uckuwya-lin].
Peter **3-mother-OBV.SG** 3-fear.TI-**APPL-3OBJ-SG** 3-come.AI-OBV
- b. Piyel Ø-nokatom-**uw-a-n** [Ø-uckuwya-lin **w-ikuwoss-ol**].
Peter 3-fear.TI-**APPL-3OBJ-SG** 3-come.AI-OBV **3-mother-OBV.SG**
Both: 'Peter is afraid of **his mother** coming.' (MA 2025.01.03)

It's also unclear what the LF requirement would be, because RtO in Passamaquoddy-Wolastoqey can also reconstruct:⁶

- (22) Reconstruction with RtO *Passamaquoddy-Wolastoqey (Eastern Algonquian)*
- a. *Context: Roger wants all of the chairs to be occupied.*
'-Pawatom-**uw-a-n** **wasis-ol** Roger ['t-opi-lin ⟨*wasisol*⟩
3-want.TI-**APPL-3OBJ-SG** **kid-OBV.SG** Roger 3-sit.AI-OBV
psi=te qotoputi-hkuk].
all=EMPH chair-LOC.PL
'Roger wants **a child** to sit in every chair.' (MA 2025.02.21) *want* >> ∀ >> ∃
- b. *Context: I'm hosting a small party, and I decide it would be nice to go on a group hike. But I have some food cooking on the stove, so someone should stay back and watch over it. I don't care who stays behind.*
Pesq pomawsuwin n-pawatom-**uw-a-n** [⟨*pesq pomawsuwin*⟩ skat
one person 1-want.TI-**APPL-3OBJ-SG** NEG
Ø-naci= yaluhsa-w-on].
3-go= walk.AI-NEG-SG
'I want **one person** to not go on the hike.' (EM 2024.12.27) *want* >> ∃ >> ¬

In (22a), *wasisol* 'a child' scopes under the downstairs quantifier *psi-te qotoputihkuk* 'every chair'. In (22b), *pesq pomawsuwin* 'one person' reconstructs to an intermediate position (e.g. Spec,TP) above embedded negation but below the intensional operator.

Since there are no PF or LF restrictions on RtO in Passamaquoddy-Wolastoqey, it must be analyzed as movement driven by a purely-syntactic feature (e.g. a movement-driving feature on Appl; see Grishin 2023, 2025 for more discussion). Thus, Meadows's (2024) attempt to explain away RtO and save the GBoIM fails here. The GBoIM undergenerates.

⁶Compare this to Nez Perce RtO, which Deal (2018) argues involves movement into a θ position and thus cannot reconstruct (or else the LF would fail to compose)—this might then plausibly be driven by conditions of LF interpretability.

4.2 Contra the Generalized A-after- $\bar{A}$ Constraint

Contra the GAa $\bar{A}$ C, (pure) $\bar{A}$ movement *can* in fact feed A operations like ϕ agreement: in languages like Passamaquoddy-Wolastoqey (Eastern Algonquian) and Tigrinya (Ethio-Semitic), downstairs $\bar{A}$ movement can feed cross-clausal object agreement.⁷

- (23) a. *Passamaquoddy-Wolastoqey (Eastern Algonquian)*
 Laca 'kociciy-**a-l** [**tehpu mahtoqehsu-wol** eli kisi=
 Roger 3-know.TA-**3OBJ-OBV.SG** **only** **rabbit-OBV.SG** IC.C PFV=
 mil-uk Ø-nican-ok *<tehpu mahtoqehsuwot>*].
 give.TA+O-1SG:3CJ 1-child-PROX.PL
 (lit.) 'Roger knows [that **only a rabbit** I gave my children *<only a rabbit>*].'
 (MA 2025.06.20)
- b. *Tigrinya (Ethio-Semitic)* (Overfelt and Cacchioli 2024:14)
 [**n=ot-a** **məts'haf** ?anə *<nəta-məts'haf>*
 ACC=**that.FSG** **book** I
 ki-Ø-nbīb-a] ji-giba?-**a**
 IRR-SM.1SG-read.IPFV-OM.3FSG SM.3MSG-need.IPFV-**OM.3FSG**
 (lit.) 'It's necessary [that **that book** I read *<that book>*].'

In (23a), downstairs focus-fronting of the ditransitive theme feeds matrix object agreement, and in (23b), downstairs topicalization of the object similarly feeds matrix object agreement. And there's abundant evidence that it's really $\bar{A}$ movement of a downstairs DP to the left edge of the complement clause that feeds matrix ϕ agreement, from diagnostics like WCO, locality, island-sensitivity, word order, and reconstruction (see citations above). The GAa $\bar{A}$ C is thus too strong, as it rules out these kinds of $\bar{A}$ -movement-fed long-distance agreement (LDA) phenomena.

5. The Ban on Improper Movement makes good predictions

The classic BoIM, in contrast, cuts the pie correctly. In Passamaquoddy-Wolastoqey and Tigrinya, you have both (hyper)RtO and LDA constructions. RtO in these languages is subject to A-locality (only the DP in the highest A position downstairs can raise), and raising can feed further A-movement (see Grishin 2025). Put differently, *only A movement can feed RtO* (and not $\bar{A}$ movement, deriving its A-locality profile), just as predicted by the BoIM. In contrast, LDA in these languages is subject to $\bar{A}$ -locality: any possible $\bar{A}$ -moving DP can in principle trigger LDA. And, strikingly, these $\bar{A}$ -moved LDA triggers *cannot then be A-moved* (Bruening 2001, Grishin 2025), exactly as the BoIM predicts.

Moreover, these kinds of facts really are about movement types and not positions/clause size, because C in some languages can bear A-movement-driving features, making Spec,CP

⁷On Passamaquoddy-Wolastoqey, see Bruening (2001), Grishin (2025); on Tigrinya, see Overfelt and Cacchioli (2024), Overfelt (2025).

an “A position”, and thus allowing for hyperraising out of CP (e.g. in Mongolian; Fong 2019, Gong 2023). Notably, even though hyperraising in these languages proceeds through Spec,CP, *it’s still subject to A-locality*—downstairs $\bar{A}$ movement cannot feed hyperraising! This indicates that what’s crucial for understanding variation and universality in cross-clausal A dependencies is the A vs. $\bar{A}$ status of each movement step (see also Lohninger 2025), not (just) positions (contra GBoIM-like approaches).

6. An additional prediction to be explored

The binding-based account of the BoIM presented here makes a very interesting prediction: given that there seem to be *base-generated* A and $\bar{A}$ binding dependencies (e.g. Hewett 2025a, a.o.), we expect to find BoIM-like effects *even in the absence of movement*, (e.g. with non-movement-derived resumption). For instance, we’d expect that a base-generated A-binder *could* A-move, but a base-generated $\bar{A}$ -binder *couldn’t*.

Notably, this prediction holds for many different variants of the kind of account presented here, even if they differ in their formal details. As long as there’s a semantic difference between A and $\bar{A}$ binding, this difference shows up in the LF interpretation of A and $\bar{A}$ movement, and we derive the BoIM from this difference, we predict BoIM-like effects even in the absence of movement. I leave it to further research to unearth relevant data that could address this prediction.

7. Conclusion

A and $\bar{A}$ movement demonstrably interact differently with (variable) binding, e.g. with WCO, depictives, and parasitic gaps. At least one theory of these differences—the kind of approach pursued by Sauerland (1998, 2004), Ruys (2000), van Urk (2015)—has the (as far as I’m aware, yet-unnoticed) property of deriving the BoIM as a consequence of the semantic differences between A and $\bar{A}$ binding. This is a good result, because we *do* want to derive the BoIM—recent alternative generalizations like the GBoIM and the GAa $\bar{A}$ C can be shown to be empirically inadequate.

This paper has merely presented a sketch of a possible binding-based account of the BoIM—there are, obviously, many loose threads to follow up on. For instance, it has often been pointed out that there are more movement types than are apparent from this binary A/ $\bar{A}$ distinction, e.g. (different types of) scrambling, *wh* vs. focus vs. relativization vs. topicalization, etc. (e.g. Müller and Sternefeld 1993, Abels 2007, 2009, a.o.). Based on (some of) this data, Abels (2007) proposes the Universal Constraint on the Ordering of Operations in Language (UCOOL), (a portion of) which is presented below, where the operations on the left must derivationally precede the operations on the right.

(24) Merge at θ positions $\prec$ A mvmt. $\prec$ scrambling $\prec$ *wh* mvmt. $\prec$ topicalization

The classic BoIM can be thought of as a portion of this larger ordering, as Abels points out. However, this ordering is essentially a stipulation that begs for an explanation (just like the original BoIM)—ideally we’d be able to derive this larger ordering from other principles.

It remains to be seen whether (a version of) the binding-based approach pursued here is up for this task (cf. Müller and Sternefeld 1993 for some early discussion of these issues). Another issue pointed out by this literature is that restrictions on the ordering of movement types don't just make predictions about the kinds of movement dependencies a *single* XP can take part in, but also for things like remnant movement, surfing/smuggling derivations, etc.—another topic to be investigated in more detail in future work.

In another vein, an anonymous reviewer points out that this proposal has nontrivial consequences for reconstruction and its interaction with BoIM effects. If reconstruction involves genuinely interpreting lower copies of moved element at LF, we might predict nontrivial interactions between reconstruction and improper movement. I have not yet explored the possible predictions this approach would make for reconstruction.

Finally, I'd like to address how this proposal fits in with recent proposals to derive BoIM(-like) effects from a *categorical* difference in A-moving vs. $\bar{A}$ -moving constituents (e.g. Safir 2019, Hewett 2025a). For instance, borrowing ideas from Cable (2010), Hewett (2025a) proposes that A movement involves movement of DPs, whereas $\bar{A}$ movement involves movement of QPs. For Hewett (2025a), this is meant to derive the GAa $\bar{A}$ C, as QPs don't bear the right kinds of features (e.g. [D], [ϕ], etc.) to undergo A movement. However, if we allow Q to bear a ϕ probe (or similar), thus acquiring A features, we'd then predict that those QPs could participate in A phenomena like ϕ agreement. As far as I can tell, then, this kind of approach is entirely compatible with what I present here (a similar observation is made by Hewett 2025a), as long as this kind of appropriate tweak is made.

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Peter Grishin
peter_grishin@brown.edu